Reboot the Earth Hackathon: Participant Briefing

UC Davis x United Nations | October 23–24, 2026

Welcome! This doc has everything you need for the hackathon. Start here, then find each challenge's full statement and data in the tabs on the left.

When and where

- **Friday, October 23:** 9am–5pm
- **Saturday, October 24:** 12–6pm
- **Where:** Shields Library, Room 167, UC Davis
- **Bring:** your laptop and charger
- **Food:** lunch, snacks, and drinks both days
- **Wi-Fi:** UC Davis guest network, Eduroam
- **Be in the room for:** the challenge announcements and team formation on Friday morning, and judging on Saturday. The rest of the time you're welcome to step outside to work
- Please use **#RTEUCDavis2026** in your social posts!

Join the Discord

We'll use Discord for everything during the event: announcements, finding a team, and questions for the organizers. Join here: https://discord.gg/t7Bb9dmg6

Once you're in, turn on notifications for #announcements (click the bell icon at the top of the channel) so you see updates from the organizers!

Teams

Every team needs 5 people. We have a large number of participants this year, so in order to keep judging running smoothly we really need full teams. On Friday, organizers will go around checking team sizes, and teams with fewer than 5 will be merged or have people added.

Start building your team as early as you can, so you're not rearranged later:

- **No team yet?** Post in #team-formation on Discord with your interests and skills
- **Team with fewer than 5?** Post there too, saying how many people you need
- **Team of 5?** Register your team!

The challenges

- **Seeing the Forest for the Trees for Fire Management:** build a better way to browse and filter the Open Forest Observatory's drone and field data. See the Fire Challenge tab.
- **Counting the Cold for California Growers:** help growers track chill and growing degree days for their own orchards, even where weather stations are far away or missing. See the Chill Hours Challenge tab.
- **Soil Intelligence (with FAO):** build AI-powered tools that help monitor and restore soil health. See the Soils Challenge tab.

Each challenge tab has a Data subtab underneath it with the datasets and how to get started. If you have any trouble, post on Discord and tag the organizers, or flag down a mentor or organizer in the room!

Judging

📍**Submit your project on Unite Ideas by 3:00pm Saturday!**
Teams will need to register for an account in order to submit.

Judging happens Saturday afternoon, in two rounds:

- **Round one:** judges walk around in small groups and visit your table. Stay at your table with your team number visible. Each visit is 3 minutes for your pitch and 2 minutes of Q&A, and every team is seen by at least two judges
- **Finals:** the top 2–3 teams across all three challenges pitch to the whole room, and a separate panel of judges picks the winners. Whether or not you're a finalist, please stay to cheer on your fellow hackers!
- **Wrap-up:** winners are announced at 5:30, followed by a group photo

Judges score each team from 1 (incomplete) to 5 (exceptional) on three things:

- **Technical execution:** does it work? Is the build solid for the time available?
- **Creativity and innovation:** is the approach original or surprisingly clever?
- **Impact:** does it address the challenge? Who benefits, and how meaningfully?

Code of conduct

Everyone at the hackathon (participants, mentors, and organizers) follows the UC OSPO Network Code of Conduct, online and in person. If someone makes you uncomfortable or breaks the code of conduct, tell any organizer in person or DM one on Discord. Reports are handled privately. Organizers: Vessela Ensberg, Pamela Reynolds, and Laura Langdon.

Helpful tools

AI tools: AI use is allowed. UC Davis students have Gemini through the university. Free options include Google AI Studio, Hugging Face open-source models, and GitHub Models.

Databases: Supabase has a free tier.

Hosting a web app: Vercel and Cloudflare Pages both have free tiers.

Preparing your pitch

These clips come from talks about pitching to investors, but the storytelling advice works for any pitch:

- Clip 1 (12:51–14:57)
- Clip 2 (6:11–7:00), then 13:20–16:33 and 30:57–31:15

Submitting your project

Submit your project on Unite Ideas by 3:00pm Saturday: Reboot the Earth - UC Davis 2026. Unite Ideas is the United Nations' platform for hackathon projects.

- **Make an account before Saturday:** Click Register at the top of the page, or log in with a Google account. This how-to walks you through it
    - Make sure to check your spam folder for the verification email!
    - If you tried signing in with Google and that doesn't seem to be working, try email instead
- **Click "Submit Your Idea":** Only the title and description are required, but add links to your code and demo too
- **Get it in by 3:00:** You can edit it as often as you like until then

Submissions are public, so anyone can see your project once it's posted.

Reboot Earth California UC Davis 2026—Fire Challenge

Seeing the Forest for the Trees: Build a Better Way to Explore Forest Drone and Field Data

Problem

Forest fire is an increasing risk globally, and land managers need tools and data to help them prioritize where to conduct fuel reduction treatments that mitigate future fire risk. The Open Forest Observatory (OFO) at UC Davis hosts detailed drone-derived forest maps across more than 600 locations in the western US. It also hosts traditional ground-based forest inventory data, in the form of mapped trees with their species, size, and live/dead status, across nearly 300 field plots. The data is open, but finding the records that match a particular place or question is still harder than it should be, and OFO is looking to redesign its web-based Data Catalog.

Challenge

Data only helps if people can find it. A land manager weighing whether an area needs forest thinning wants to know what drone and field data exists nearby, how it was collected, and whether they're allowed to use it. Someone quantifying how well thinning has been used to reduce tree density needs to find the plots and drone flights that let them make that comparison.

This challenge invites participants to build a web interface that helps users browse and filter OFO data records based on their spatial footprints and metadata attributes. Teams work from a lightweight catalog of three files, about 12 MB in total, describing every drone mission, ground plot, and mapped tree, each with a metadata file explaining its columns. The task is about the design and engineering of the interface, with no prediction or model training involved. Strong solutions may be incorporated into a redesign of the OFO Data Catalog (https://openforestobservatory.org/data).

Participants may build map-based search, filters on attributes such as measurement protocol, minimum tree size, species, or license, views that show which other records sit near a selected drone mission or ground plot, ways to see which drone missions cover which ground plots, and previews that link out to the drone imagery itself. Everything needed to get started is in the Data tab.

Fire Challenge Data

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Everything you need for the “Seeing the Forest for the Trees: Build a Better Way to Explore Forest Drone and Field Data” challenge is in one Zenodo record (Zenodo is a free, open repository for research data): **https://zenodo.org/records/23402904**

If you have any trouble, post on Discord and tag the organizers, or flag down a mentor or organizer in the room!

1. Read the record's description first

The description on the Zenodo page explains what the data is, how it's licensed, and what to watch out for. The data team asks that everyone read it in full before using the data. It takes about ten minutes.

2. Download the files

Click **Download all** on the Zenodo page, or grab the files one by one. It's about 12 MB in total, so it's quick. There are three data files, each with a metadata file that explains every column:

Data file	What each row is	Read alongside
`ofo_ground-reference_plots.gpkg`	One **ground plot**: an area where field crews measured trees by hand (about 300 plots)	`ofo_ground-reference_plots_metadata.md`

Data file	What each row is	Read alongside
`ofo_ground-reference_trees.gpkg`	One **tree**, as a point, with its species, size, and live/dead status	`ofo_ground-reference_trees_metadata.md`
`ofo_drone-missions.gpkg`	One **drone mission**: the area a drone flight covered (600+ locations)	`ofo_drone-missions_metadata.md`

Trees link to plots through the `plot_id` column.

3. Open the data

The `.gpkg` files are GeoPackages, a standard format for map data: each row is a shape on a map (a plot or mission footprint, or a tree's location) plus columns describing it. Pick whichever tool you're comfortable with:

- **Python:** `geopandas.read_file("ofo_ground-reference_plots.gpkg")`
- **R:** `sf::st_read("ofo_ground-reference_plots.gpkg")`
- **Point and click:** drag the file into QGIS (free) to see it on a map
- **For a web app:** most web map libraries (Leaflet, MapLibre, deck.gl) want GeoJSON. Convert with `gdf.to_crs(4326).to_file("plots.geojson", driver="GeoJSON")` in geopandas, or `ogr2ogr -f GeoJSON -t_srs EPSG:4326 plots.geojson ofo_ground-reference_plots.gpkg` on the command line

4. Look at what exists today

Your challenge is to build a better way to browse and filter these records. The current Open Forest Observatory catalog is at https://openforestobservatory.org/data/. Spend a few minutes with it so you know what you're improving on. Look at both the drone data and the ground-reference data.

5. The drone imagery itself (optional)

The files you downloaded in step 2 are the catalog: they describe each plot, tree, and drone mission. The drone imagery itself, and the maps made from it (orthomosaics, canopy height models, 3D point clouds), is stored separately in a public cloud bucket. These files are large, and you don't need to download them to build a catalog interface. If you want to show image previews in your interface, link to the files rather than downloading them. The Zenodo description gives the URL pattern, and a sample file listing is at https://js2.jetstream-cloud.org:8001/swift/v1/ofo-public/.

Things worth knowing

- **Each record has its own license.** Plots and missions come from many contributors, and each carries its own Creative Commons license in the `license` column. Your interface should show it, and anything you share should credit the contributors
- **Plots weren't all measured the same way.** Different plots used different minimum tree sizes and protocols. The plots metadata file explains the columns that capture this, and they're good candidates for filters
- **Plots and drone missions don't match one-to-one.** A plot can be covered by several missions or none, and vice versa. Ground plots can also sit up to about 10 m offset from the drone imagery (due to GPS position error when surveying the field plot)
- **Many locations have two drone missions:** a high-altitude straight-down flight and a low-altitude angled one
- **`withhold_from_training`** marks about 20% of records reserved for testing machine-learning models. This challenge doesn't involve model training, but it's a field users would want to see

Initial ideas

- A means of filtering data records would likely be a very useful functionality.
- When viewing a single record (e.g. a drone mission or a ground plot), it may be helpful for the user to be able to see what other records are nearby.

Questions

Derek Young from the Open Forest Observatory is mentoring both days, and is the expert on this challenge!

Reboot Earth California UC Davis 2026—Chill Hours Challenge

Counting the Cold: Help California Growers Track Chill and Growing Degree Days

A challenge from Julia Stover of the UC Davis Fruit and Nut Research and Information Center (FNRIC).

Problem

Fruit and nut growers across California plan their seasons around two numbers calculated from air temperature. Chill measures the winter cold a tree has had, which tells growers whether bud break is likely to be on time and even, and whether newly opening buds may need frost protection in a cold spring. Growing degree days (GDD) add up heat over the season and drive predictions of everything from pest and disease life cycles to harvest times. FNRIC's models rely mainly on CIMIS, California's statewide network of agricultural weather stations. But in regions with rapid elevation changes, the nearest station may be in a very different microclimate from the orchard, and in some large agricultural regions stations have been removed and not replaced, leaving growers many miles from the nearest one.

Challenge

A grower wants to know how their own orchard is doing this season, not how a weather station many miles away is doing. Their questions are simple: How much chill or GDD have I accumulated so far? How does that compare to past years? How much should I expect in the next week, or the next two?

This challenge invites participants to build a tool that answers those questions for a grower's location, reporting chill, GDD, or both, accumulated so far or projected to a certain date. FNRIC's models show how the calculations work, and teams can draw on CIMIS and

other weather station networks for the data. Evapotranspiration (ET), the water-use model growers also rely on, is out of scope for this hackathon. You don't have to cover the whole state: even a few locations in the Central Valley would make a valuable tool.

Participants may build dashboards or apps that show this season's chill and GDD alongside past years, short-range forecasts of chill and GDD, ways to estimate conditions at an orchard far from a station by combining several station networks or gridded weather data, links from chill and GDD to a crop model such as FNRIC's almond hull split model, or an open API that runs the calculations for several models at once. Everything needed to get started is in the Data tab.

Chill Hours Challenge Data

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The data for the "Counting the Cold: Help California Growers Track Chill and Growing Degree Days" challenge comes from CIMIS, the California Irrigation Management Information System: a statewide network of agricultural weather stations run by the California Department of Water Resources. The stations below have records going back to the 1980s, so you can compare this season with decades of past ones.

If you have any trouble, post on Discord and tag the organizers, or flag down a mentor or organizer in the room!

1. Get a CIMIS app key

The CIMIS Web API needs an app key. Register for an account at cimis.water.ca.gov (use “New CIMIS”), and when the form asks for a phone number, type digits only, with no parentheses or dashes. Once you verify your email, your app key will be on your account page. Do this first, since one person can share the key with the whole team.

2. Pick your stations

You don't need the whole state. These five Central Valley stations, from north to south, are all active and have records back to the 1980s:

Station number	Name	County
6	Davis	Yolo
70	Manteca	San Joaquin
71	Modesto	Stanislaus

Station number	Name	County
39	Parlier	Fresno
5	Shafter	Kern

For every station, with its location, elevation, and whether it's still active, see the station list (no app key needed): https://et.water.ca.gov/api/station

3. Download the data

Each request names the stations, the dates, and the data items you want. For example, this request gets hourly air temperature in °F for Davis over one winter:

```
https://et.water.ca.gov/api/data?appKey=YOUR-APP-KEY&targets=6&startDate=2025-11-01&endDate=2026-02-28&dataItems=hly-air-tmp&unitOfMeasure=E
```

The data items you'll need:

- `hly-air-tmp`: hourly air temperature, for chill
- `day-air-tmp-max` and `day-air-tmp-min`: daily high and low, for GDD

Ask for one station and one season at a time. Very large requests are refused.

4. Calculate chill and GDD

The simplest versions, in Python:

- **Chill hours:** count the hours between 32°F and 45°F. `chill_hours = ((temps >= 32) & (temps <= 45)).sum()`
- **GDD:** for each day, the day's average temperature above a base temperature, added up over the season. `gdd = max(0, (tmax + tmin) / 2 - base)`

FNRIC also uses other chill models (chill units and chill portions), and each crop has its own GDD base temperature. FNRIC explains the three ways of counting chill and has all its weather models in one place.

5. Look at what exists today

Spend a few minutes with these so you know what you're building on:

- FNRIC's chill calculator
- UC ANR's future chill portions tool, for long-term chill projections

Things worth knowing

- **Pick a start date.** Chill and GDD are running totals, so they mean little without a date to start counting from. Say which one your tool uses
- **Missing hours happen.** Stations go offline and readings get flagged. Check for gaps before adding up hours, and decide how your tool handles them
- **Stations close.** Durham (station 12), in Butte County's orchard country, stopped reporting in July 2025. The station list shows which stations are inactive
- **Not every station suits farming.** Even CIMIS stations aren't always well maintained, and other networks' stations, like those at airports, often sit near concrete or on tall towers

Additional resources

You don't need these to get started, but they may help if you take your tool further:

- **Wildeye station data:** FNRIC's network of in-orchard weather stations, funded by the pistachio industry
- **PRISM:** free daily temperature estimates for the whole US at 800 m resolution, for orchards far from any station
- **FNRIC Data Primer and Resources:** Everything you need to pull California chill and growing degree day (GDD) data from the Fruit & Nut Research & Information Center (FNRIC), check your math against it, and extend it with a forecast.

Questions

Julia Stover from FNRIC will stop by Friday morning, and Andy Lyons from UC ANR is mentoring all day Friday.

Background from FNRIC

Julia Stover's full framing of the problem:

Weather data is critical for California growers, supporting both short and long-term management decisions.

Chill models rely on hourly air temperature data and are used by growers of most deciduous fruit and nut tree crops to determine whether they are likely to experience difficulties with timing and uniformity of bud break, which can lead to issues with fruit set and quality later in the season. In cold springs, it can predict whether frost protection measures are likely to be needed for newly opening buds. Longer-term, looking at annual chill patterns can inform decisions about variety selection.

Growing degree day (GDD) models are based on daily max and min air temperature measurements and are used by growers of almost all crops for everything from prediction of pest and disease life cycles to harvest times. Longer-term, GDD patterns support orchard establishment decisions.

Evapotranspiration (ET) models describe the water lost to the atmosphere through evaporation and transpiration. Calculation of ET requires measurements of solar radiation, air temperature, relative humidity, and wind speed. However, for precision, most ET is calculated from reference crop ET. In most cases, this means it requires a standardized turf surface below the weather station. ET is used to make irrigation scheduling decisions to meet water use regulations and conserve water while avoiding drought stress levels that would negatively impact yield or crop quality. Longer-term, these patterns can help growers select more resilient varieties or crops as they plan for increasing climate variability.

FNRIC models rely mainly on CIMIS stations for weather data, as they are widely available throughout the state, and provide the data necessary for all three of these models. However, CIMIS has significant limitations. In regions with rapid elevation changes, a nearby CIMIS station may be in a very different micro-climate than neighboring orchards. In some large agricultural regions, CIMIS stations have been removed and not replaced, leaving growers many miles from their nearest station. In-orchard weather stations are sold inexpensively by many different companies, and they are generally believed to be accurate for air temperature measurements. However, hourly measurements can be difficult to transmit in remote areas, and irrigated turf is required for the calculation of ETo. Developing techniques for predicting and extrapolating data from more sources will be a critical tool as growers rely more and more on weather and production models to navigate a changing climate.

Reboot Earth California UC Davis 2026—Soil Challenge

AI for Regenerative Landscapes and Soil Intelligence

A challenge from the Food and Agriculture Organization of the United Nations (FAO).

Problem

Healthy soils are fundamental to food production, climate resilience, biodiversity conservation, and carbon sequestration. However, soil degradation, declining organic matter, erosion, and unsustainable land management practices continue to threaten productivity across many agricultural regions. Monitoring and restoring soil health at scale remains a significant challenge.

Challenge

The future of agriculture depends on restoring ecosystems while maintaining productivity and profitability. Better intelligence is needed to understand soil conditions, measure environmental impacts, and support regenerative agricultural transitions.

This challenge invites participants to create AI-powered Soil and Landscape Intelligence Solutions that enhance soil health, support carbon-positive agriculture, and strengthen ecosystem resilience.

Participants may develop soil diagnostics platforms, carbon measurement and verification systems, regenerative agriculture advisory tools, ecosystem monitoring solutions, biodiversity assessment technologies, digital soil marketplaces, and climate-smart farm management platforms that accelerate the transition toward sustainable and regenerative landscapes. Everything needed to get started is in the Data tab.

Soil Challenge Data

Reboot the Earth Hackathon
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The data for the "AI for Regenerative Landscapes and Soil Intelligence" challenge comes in two parts, and no account or app key is needed for either:

- **Soil properties** from the California Soil Resource Lab at UC Davis: maps of more than 50 soil properties, from organic matter and pH to water-holding capacity and erosion risk. Each map is a grid of 800 m squares covering the continental US, built from current USDA soil surveys
- **Soil carbon** from FAO's Global Soil Organic Carbon Sequestration Potential Map (GSOCseq): how much carbon California's farmland holds today, and how much it could hold after 20 years of better soil management. We've cut the global maps down to California for you

If you have any trouble, post on Discord and tag the organizers, or flag down a mentor or organizer in the room!

1. Explore the maps first

Open the Soil Properties app, pick a property, and zoom in on California. Each property's metadata in the app explains what it measures, its units, and how it was calculated. Five minutes here will help you choose what to download.

2. Pick your soil properties

You don't need all of them. Each property is its own file, and most are about 50 MB. These are a good place to start:

Property	File	Units	Why it matters
Soil organic matter	om_kg_sq_m.tif	kg/m²	The main store of carbon in soil, and a core measure of soil health
pH	ph.tif	pH	Affects which nutrients plants can take up
Electrical conductivity	ec.tif	dS/m	Salinity, a growing problem in irrigated land
Clay	clay.tif	percent by weight	Shapes how soil holds water and nutrients
Available water holding capacity	water_storage.tif	cm	How much water the soil can store for plants
Erodibility (0–25 cm)	kw_025.tif	factor value	How easily water washes the topsoil away
Land capability class, irrigated	i_class.tif	class	How well suited the land is to farming

The full list of 54 properties is on the download page.

3. Get the carbon maps

Download the GSOCseq California folder from Google Drive. It's six files, about 17 MB in total, plus a README:

File	What each cell holds
GSOCseq_T0_Map030_CA.tif	Soil organic carbon in the top 30 cm in 2020, the starting point, in tonnes of carbon per hectare

File	What each cell holds
GSOCseq_finalSOC_BAU_Map030_CA.tif	Carbon after 20 years if management stays the same ("business as usual")
GSOCseq_finalSOC_SSM1_Map030_CA.tif	Carbon after 20 years of sustainable soil management that adds 5% more carbon to the soil
GSOCseq_finalSOC_SSM2_Map030_CA.tif	The same, adding 10% more
GSOCseq_finalSOC_SSM3_Map030_CA.tif	The same, adding 20% more
GSOCseq_T0_UncertaintyMap030_CA.tif	How uncertain the starting estimate is

Subtract the business-as-usual map from a sustainable management map to see how much extra carbon better management could store in each place.

4. Download the soil property files

Click Download next to each property on the download page. Every file follows the same pattern, so you can also fetch them from the command line:

curl -O https://casoilresource.lawr.ucdavis.edu/800m_grids/rasters/om_kg_sq_m.tif

Download only the properties you need, and have one person download and share them with the team, so the venue Wi-Fi holds up.

5. Open the data

All the files are GeoTIFFs, a standard format for map grids: each cell is one square with one value. The carbon maps are ready to use. They're already cut to California, they use latitude and longitude, and their cells are about 1 km across. The soil property files cover the whole continental US at 800 m, so clip them to California (or your own area) first. Pick whichever tool you're comfortable with:

- **Python:** rioxarray.open_rasterio("om_kg_sq_m.tif").rio.clip_box(-124.5, 32.5, -114, 42, crs="EPSG:4326")
- **R:** r <- terra::rast("om_kg_sq_m.tif"), then terra::crop(r, terra::project(terra::ext(-124.5, -114, 32.5, 42), "EPSG:4326", terra::crs(r)))
- **Point and click:** drag the file into QGIS (free) to see it on a map
- **For a web app:** most web map libraries want latitude and longitude. Clip and convert in one step with gdalwarp -t_srs EPSG:4326 -te -124.5 32.5 -114 42 -te_srs EPSG:4326 om_kg_sq_m.tif om_ca.tif on the command line

6. Look at what exists today

Spend a few minutes with these so you know the current context:

- SoilWeb: the lab's map of detailed soil survey data, down to individual fields
- The Soil Properties app from step 1

Things worth knowing

- **These maps are coarse.** They're made to show regional patterns, and the lab notes that a cell's value may not be accurate at a specific location. They're good for "where" questions, not for replacing one farm's soil test
- **They're one snapshot.** The soil property maps show soil as surveyed, not how it changes year to year, so they can't measure whether a practice is working. The carbon maps are model estimates of what could happen, not measurements
- **The two datasets use different map projections.** The soil property grids use a US-wide projection (Albers Equal Area, EPSG:5070, in meters); the carbon maps use latitude and longitude. Reproject the soil grids before combining them with the carbon maps, as in step 5
- **The carbon maps only cover farmland.** Cells outside agricultural land have the value -999, which means no data
- **The carbon maps are for non-commercial use.** They're under a CC BY-NC-SA 3.0 IGO license: credit FAO, don't use them in anything you sell, and share anything you build from them under the same license

- **Some properties are categories, not numbers.** Drainage class, soil order, texture, and hydrologic group are stored as integer codes; the property's metadata in the app has the lookup table
- **Depth matters.** Some properties come in several depth ranges (0–5 cm, 0–25 cm, and so on), named in the file. Check the metadata for what depth the others cover, and say which depth your tool uses
- **Check for empty cells.** Some cells have no value, so decide how your tool handles them before averaging or modeling

Additional resources

You don't need these to get started, but they may help if you take your tool further:

- Soil Data Access: USDA's full soil survey database, for field-level detail behind the 800 m grids
- Tools / Apps from the CA Farm Demonstration Network (including two soils apps)
    - https://www.calfarmdemo.org/find-a-tool/
    - Sample CA Soil Health Report
- DWR Crop Distribution Layer
    - Sample R code
- DWR Well Data
- CropManage - decision support tool for nutrient and irrigation

Initial ideas

- A soil health scorecard for any location in California: which properties hold it back (salinity, low organic matter, erosion risk) and what practices could help
- An advisory assistant that answers a grower's questions in plain language, grounded in the soil data for their location
- A map of where better management would store the most extra carbon, alongside the soil properties that could hold it back
- A carbon estimator for a farm: today's soil carbon, what it could be in 20 years under each scenario, and how confident the estimate is

Questions

Tony Lei is mentoring this challenge Friday from 10am to 2pm.